

MCQ

Course Title: Co-ordinate Geometry and Vector Analysis

Course Code: MATH 2101

Q. The equation of the X-axis is:

- A) $x = 0$
- B) $y = 0$
- C) $y = x$
- D) $x + y = 0$
- E) $xy = 1$

Q. The coordinates of a point are given as $(a, 0)$. This point must be on:

- A) X-axis
- B) Y-axis
- C) Origin
- D) Line $y = x$
- E) A diagonal

Q. If the equation of a line is $y = 3x + 5$, what is its y-intercept?

- A) -5
- B) -3
- C) 1
- D) 3
- E) 5

Q. The straight line equation $y = 5x - 2$ has a slope of-

- a) 5
- b) 2
- c) $x + y$
- d) $x - y$
- e) xy

Q. A pair of equations to determine the value of two variable is called-

- a) Simultaneous linear equation
- b) paired equation
- c) quadratic equation
- d) simple equation
- e) differential equation

Q. Line integral is used to calculate.....

- a) force
- b) area
- c) volume
- d) length
- e) mass

Q. Surface integral is used to compute.....

- a) force
- b) area
- c) volume
- d) length
- e) mass

- Q. If the sum of the distance of a point from two perpendicular lines in a plane is one, then its locus is.....
- a) A circle
 - b) Straight line
 - c) Two intersecting line
 - d) ellipse
 - e) **Square**
- Q. The surface integral of the electric field intensity is.....
- a) electric charge
 - b) volume flux
 - c) **net flux emanating from the surface**
 - d) area of flux
 - e) net electric charge
- Q. The divergence theorem converts.....
- a) Line to surface integral
 - b) **Surface to volume integral**
 - c) Volume to line integral
 - d) Surface to line integral
 - e) Volume to surface integral
- Q. The mathematical perception of the gradient is said to be
- a) Chord
 - b) tangent
 - c) **slope**
 - d) arc
 - e) vector
- Q. Which transformation is **not** distance-preserving?
- A) Reflection
 - B) Rotation
 - C) Translation
 - D) **Scaling**
 - E) differentiation
- Q. What is the equation of the circle $x^2 + y^2 = 25$, when the origin is shifted to the point $(3, -4)$?
- A) $(x - 3)^2 + (y + 4)^2 = 25$
 - B) **$(x + 3)^2 + (y - 4)^2 = 25$**
 - C) $(x + 3)^2 + (y + 4)^2 = 25$
 - D) $(x - 3)^2 + (y - 4)^2 = 25$
 - E) $x^2 + y^2 = 25$
- Q. If two lines are perpendicular, the sum of the product of their **direction cosines** is:
- A) -2
 - B) -1
 - C) **0**
 - D) 1
 - E) 2
- Q. Line integral is used to calculate.....
- a) force
 - b) area
 - c) volume
 - d) **length**
 - e) mass

Q. In a plane, the direction of a vector is denoted by

- a) line
- b) point
- c) angle
- d) arrowhead
- e) force

Q. The divergence of the vector $yi + zj + xk$ is.....

- a) 0
- b) 1
- c) 2
- d) 3
- e) 9

Q. The Stoke's theorem uses which of the following operation?

- a) Gradient
- b) Divergence
- c) Curl
- d) vector
- e) Laplacian

Q. Mathematically, the functions in Green's theorem will be.....

- a) Continuous derivatives
- b) Discrete derivatives
- c) Continuous partial derivatives
- d) Discrete partial derivatives
- e) discontinuous partial derivatives

Q. Find the value of Green's theorem for $F = x^2$ and $G = y^2$ is

- a) 1
- b) 2
- c) 4
- d) 9
- e) 0

Q. The divergence theorem for a surface consisting of a sphere is computed in which coordinate system?

- a) cartesian
- b) rectangular
- c) cylindrical
- d) spherical
- e) Depends on the function

Q. The equation of a line with slope m and y-intercept c is:

- A) $y = mx - c$
- B) $y = \frac{c}{m} + x$
- C) $y = \frac{m}{c} - x$
- D) $y = c - mx$
- E) $y = mx + c$

- Q. The equation of the **YZ-plane** is:
- A) $x = 0$
 - B) $y = 0$
 - C) $z = 0$
 - D) $x + y + z = 0$
 - E) $xy = 0$
- Q. The perpendicular distance from point (1,2,3) to the plane $3x - 4y + 12z - 6 = 0$ is given by:
- A) $\frac{13}{25}$
 - B) $\frac{25}{13}$
 - C) $\frac{27}{13}$
 - D) $\frac{29}{13}$
 - E) $\frac{22}{13}$
- Q. Divergence operation result will always be.....
- a) positive
 - b) negative
 - c) scalar or vector
 - d) scalar
 - e) vector
- Q. If two functions A and B are discrete, their Green's value for a region of circle of radius a in the positive quadrant is
- a) $-\infty$
 - b) ∞
 - c) zero
 - d) one
 - e) does not exist
- Q. The point where the X-axis and Y-axis intersect is called the:
- A) Slope
 - B) Midpoint
 - C) Quadrant
 - D) Origin
 - E) Distance
- Q. If a point lies in the second quadrant, which of the following is **true**?
- A) Both coordinates are positive
 - B) Both coordinates are negative
 - C) The x-coordinate is positive, and the y-coordinate is negative
 - D) The x-coordinate is negative, and the y-coordinate is positive
 - E) The x-coordinate is zero
- Q. Which of the following points lies on the line $y = 2x - 1$?
- A) (1, 1)
 - B) (2, 3)
 - C) (0, -1)
 - D) (3, 5)
 - E) (-1, 2)

Q. What is the value of P for which the point $(-1, 3)$, $(2, p)$ and $(5, -1)$ are collinear?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

Q. If the area lies on the right, the result will be -

- a) positive
- b) negative
- c) zero
- d) one
- e) infinity

Q. If the direction cosines of a line are $(\frac{k}{3}, \frac{k}{3}, \frac{k}{3})$ then the value of k is

- a) $\sqrt{3}$
- b) $k > 0$
- c) $0 < k < 1$
- d) 3
- e) $\frac{1}{3}$

Q. When a line is parallel to a plane, the projection of the line on to that plane will be its ----- length.

- a) shortened
- b) enlarged
- c) point
- d) maximum
- e) true

Q. Which of the following is **NOT** a type of coordinate transformation?

- A) Translation
- B) Rotation
- C) Reflection
- D) Differentiation
- E) Scaling

Q. If the axes are rotated **without translation**, what property of a point remains unchanged?

- A) X-coordinate
- B) Y-coordinate
- C) Distance from origin
- D) Angle with the X-axis
- E) Equation of the line

Q. What do the **direction cosines** of a line represent?

- A) The angles the line makes with the coordinate axes
- B) The distance of a point from the origin
- C) The midpoint of the line segment
- D) The equation of the line
- E) The slope of the line

Q. Two vectors $\vec{A}$ and $\vec{B}$ will be perpendicular if

- a) $\vec{A} \cdot \vec{B} = 1$
- b) $\vec{A} \cdot \vec{B} = 0$
- c) $\vec{A} \times \vec{B} = -1$
- d) $\vec{A} \times \vec{B} = 0$
- e) $\vec{A} \times \vec{B} = 1$

Q. The Cartesian co-ordinates of the polar co-ordinate $(2, \frac{\pi}{4})$ is-

- a) $(\sqrt{2}, \sqrt{2})$
- b) $(\sqrt{2}, -\sqrt{2})$
- c) $(-\sqrt{2}, \sqrt{2})$
- d) $(2, 2)$
- e) $(\sqrt{2}, 2)$

Q. The sum of the squares of the direction cosines of every line is-

- a) one
- B) two
- c) infinity
- d) zero
- e) three

Q. If a line passes through a point $(0, c)$ and has gradient 'm' then the equation will be-

- a) $y = mx + c$
- b) $y = mx - c$
- c) $y = mx$
- d) $y = c$
- e) $y = m + c$

Q. If the area lies on the right, the result will be -

- a) positive
- b) one
- c) negative
- d) zero
- e) infinity

Q. The perpendicular distance from point $(2, 3)$ to the line $4x - 3y + 5 = 0$ is given by:

- A) $\frac{5}{4}$
- B) $\frac{4}{5}$
- C) $\frac{3}{4}$
- D) $\frac{4}{3}$
- E) $\frac{2}{3}$

- Q. If two planes are parallel, then their normal vectors are:
- A) Perpendicular
 - B) **Parallel**
 - C) Equal
 - D) Negative reciprocals
 - E) Intersecting
- Q. Which of the following is **NOT true** about direction cosines?
- A) They represent the cosines of angles with coordinate axes
 - B) They satisfy $l^2 + m^2 + n^2 = 1$
 - C) **They can be greater than 1**
 - D) They can be negative
 - E) They are unique for a given direction

Q. The gradient of $xi + yj + zk$ is

- a) 0
- b) 1
- c) 2
- d) 3**
- e) 4

Q. Which of the following theorem use the curl operation?

- a) Gauss Divergence theorem
- b) Green's theorem
- c) stocke's theorem**
- d) Maxwell equation
- e) Laplace transform